

BIOINFORMATICS MINI-PROJECT 2

Comparative Structural Analysis — Hen Egg-White Lysozyme: 1FDL vs 1AKI

1. Scientific question

Does hen egg-white lysozyme retain a broadly similar 3D fold when it is free compared with when it is bound to an antibody?

2. Structures compared

Feature	1AKI	1FDL
Context	Lysozyme alone	Lysozyme bound to Fab D1.3
Chain used	A	C [auth Y]
Length	129 residues	129 residues
Method	X-ray diffraction	X-ray diffraction
Resolution	1.50 Å	2.50 Å
Organism	Gallus gallus	Gallus gallus

3. Method

I opened both structures in Mol*, inspected the lysozyme chains and compared sequence landmarks and overall 3D appearance. The 1AKI sequence began with KVFGRCELAA and the lysozyme in 1FDL showed the same sequence context. Both records contain a 129-residue lysozyme chain.

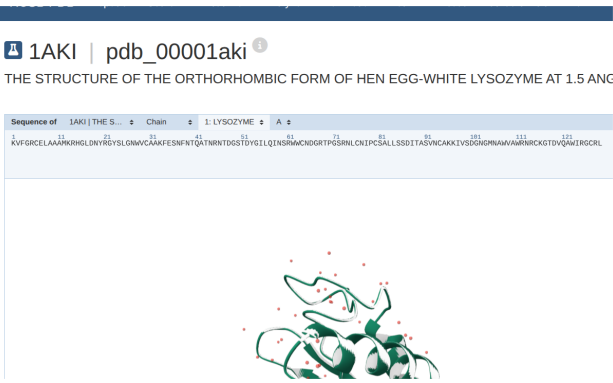


Figure 1. Clean crop of the free lysozyme structure in 1AKI.

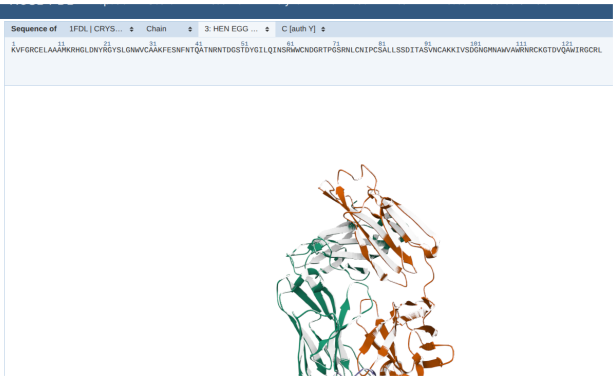


Figure 2. Clean crop of the antibody-bound lysozyme context in 1FDL.

4. Observation

The overall lysozyme shape appeared broadly similar in the free and antibody-bound structures. This is a qualitative observation, not a claim that every atom occupies the same position.

5. Resolution

1AKI has a reported resolution of 1.50 Å, while 1FDL has 2.50 Å. For X-ray crystallography, a smaller Å value generally corresponds to finer reported structural detail.

6. Interpretation and next step

The comparison suggested that lysozyme keeps its overall fold in the two biological contexts. A structural alignment and metrics such as RMSD and TM-score are needed to measure the similarity quantitatively. That became the focus of Project 4.

Conclusion

The two structures show a broadly similar overall lysozyme fold despite their different contexts. This project established the qualitative foundation for the later quantitative comparison.